



4181 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Cycle: 2, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Kilonova Sequence				
	1	Epoch 1 -- NIRSPEC spectroscopy	NIRSpec Fixed Slit Spectroscopy	(1) KILONOVA
	2	Epoch 1 -- MIRI Spectroscopy	MIRI Low Resolution Spectroscopy	(1) KILONOVA
	3	Epoch 2 -- NIRSPEC spectroscopy	NIRSpec Fixed Slit Spectroscopy	(1) KILONOVA
	4	Epoch 2 -- MIRI Spectroscopy	MIRI Low Resolution Spectroscopy	(1) KILONOVA
	5	Epoch 3 -- NIRSPEC low-res spectroscopy	NIRSpec Fixed Slit Spectroscopy	(1) KILONOVA
	6	Epoch 3 -- MIRI Imaging	MIRI Imaging	(1) KILONOVA

ABSTRACT

This proposal seeks to leverage the transformative capabilities of JWST to produce a high-quality spectroscopic dataset for one optical counterpart (kilonova) associated with a neutron star merger (NS-NS or NS-BH) detected by Advanced LIGO/Virgo/KAGRA in their observing run O4. The observations consist of three epochs from 5 to 20 days after the merger: (1) and (2) NIRSPEC + MIRI spectroscopy; and (3) NIRSPEC spectroscopy + MIRI imaging.

These observations will: (i) obtain spectroscopy of a sufficiently high SNR and broad wavelength coverage to detect and identify as many spectral features as possible; (ii) measure the ejection velocities in the different ejecta components, and (iii) precisely measure the infrared spectral energy distribution into the mid-IR to constrain the potential presence of emission components beyond those dominating in the near-IR.

These data will detect and characterize the emission from multiple ejecta components predicted by numerical simulations of neutron star mergers as part of a multimessenger investigation of the merger process. We waive our proprietary rights to enable the whole community to benefit from these data.

OBSERVING DESCRIPTION

This proposal will obtain three epochs of observations of the optical counterpart to a compact object merger containing a neutron star (NS-NS or NS-BH), detected as a gravitational wave source by LIGO/Virgo/Kagra during run O4.

There are six observation sequences, arranged into three epochs:

JWST Proposal 4181 (Created: Monday, July 1, 2024 at 6:00:46 PM Eastern Standard Time) - Overview

Epoch 1 (as soon as possible after the merger, 5 or fewer days after merger)

This requires a disruptive ToO.

Two components grouped together:

NIRSPEC medium-resolution spectroscopy with the S200A1 fixed slit and G140M/F100LP, G235M/F170LP, and G395M/F290LP grisms

MIRI low-resolution slit spectroscopy

Epoch 2 (~10 days after merger)

4-6 days after the first epoch.

Two components grouped together:

NIRSPEC medium-resolution spectroscopy with the S200A1 fixed slit and G140M/F100LP, G235M/F170LP, and G395M/F290LP grisms

MIRI low-resolution slit spectroscopy

Epoch 3 (~20 days after merger)

8-12 days after the second epoch

NIRSPEC low-resolution slit spectroscopy with the S200A1 fixed slit and PRISM

MIRI imaging with F560W and F770W filters

Proposal 4181 - Targets - Infrared Spectroscopy of a Neutron Star Merger with JWST

Generic Targets	#	Name	Criteria	Description
	(1)	KILONOVA	LVK gravitational wave source believed to contain at least one neutron star (i.e., NS-NS or NS-BH) with a confirmed optical or infrared counterpart brighter than $m \sim 23$ AB mag	
	(2)	TACQ-STAR	placeholder for a bright star ($m < 18$ AB in K) near KILONOVA to be used for target acquisition onto slits for spectroscopy	

Proposal 4181 - Observation 1 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Observation	Proposal 4181, Observation 1: Epoch 1 -- NIRSPEC spectroscopy										Mon Jul 01 23:00:46 GMT 2024	
	Diagnostic Status: Error											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Epoch 1 -- NIRSPEC spectroscopy (Obs 1)) Error (Form): A response time less than 3.0 Days is significantly disruptive to the scheduling process and is not allowed as a Carry-Over ToO.											
	(Epoch 1 -- NIRSPEC spectroscopy (Obs 1)) Warning (Form): Response time < 3.0 Days is significantly disruptive to the scheduling process. An additional direct scheduling overhead of 30.0 Mins will be added to the total time calculated for the observation.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria		Description							
	(1)	KILONOVA	LVK gravitational wave source believed to contain at least one neutron star (i.e., NS-NS or NS-BH) with a confirmed optical or infrared counterpart brighter than m~23 AB mag									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	2 TACQ-STAR	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	146079.2	
Template	Slit				Subarray							
	S200A1				FULL							
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140M/F100LP	S200A1	NRSIRS2RAPID	40	1	1	NONE	4	4	2392.578	146079.11
	2	G235M/F170LP	S200A1	NRSIRS2RAPID	25	1	2	NONE	4	4	1517.245	146079.1
	3	G395M/F290LP	S200A1	NRSIRS2RAPID	40	1	3	NONE	4	4	2392.578	146079.12

Proposal 4181 - Observation 1 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Special Requirements

On Hold Awaiting trigger from LVK detection of binary merger containing a neutron star
Target Of Opportunity Response Time 2 Days, Carry-Over

3 After 1 by 4 Days to 6 Days
Group Observations 1, 2, Non-interruptible

Proposal 4181 - Observation 2 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Observation	Proposal 4181, Observation 2: Epoch 1 -- MIRI Spectroscopy Diagnostic Status: Warning Observing Template: MIRI Low Resolution Spectroscopy								Mon Jul 01 23:00:46 GMT 2024
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Generic Targets	#	Name	Criteria	Description					
	(1)	KILONOVA	LVK gravitational wave source believed to contain at least one neutron star (i.e., NS-NS or NS-BH) with a confirmed optical or infrared counterpart brighter than m~23 AB mag						
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	2 TACQ-STAR	F560W	FAST	10	1	1	27.75	146079.33
Template	Subarray				Obtain Verification Image?				
	FULL				false				
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps	
	1	ALONG SLIT NOD							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	80	20	40	1	2	8985.58	146079.3

Proposal 4181 - Observation 2 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Special Requirements	On Hold Awaiting a GW trigger, to be scheduled at same time as NIRSPEC observations Group Observations 1, 2, Non-interruptible
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Proposal 4181 - Observation 3 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Observation	Proposal 4181, Observation 3: Epoch 2 -- NIRSPEC spectroscopy										Mon Jul 01 23:00:46 GMT 2024	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria									Description
	(1)	KILONOVA	LVK gravitational wave source believed to contain at least one neutron star (i.e., NS-NS or NS-BH) with a confirmed optical or infrared counterpart brighter than m~23 AB mag									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	2 TACQ-STAR	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	146079.2	
Template	Slit					Subarray						
	S200A1					SUBS200A1						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140M/F100LP	S200A1	NRSRAPID	360	2	1	NONE	4	8	4499.668	146079.34
	2	G235M/F170LP	S200A1	NRSRAPID	240	2	2	NONE	4	8	3003.988	146079.35
	3	G395M/F290LP	S200A1	NRSRAPID	400	2	3	NONE	4	8	4998.228	146079.13

Proposal 4181 - Observation 3 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Special Requirements	On Hold Awaiting GW trigger, to be scheduled ~5 days after Epoch 1 NIRSPEC+MIRI observations
	3 After 1 by 4 Days to 6 Days
	5 After 3 by 8 Days to 12 Days
	Group Observations 3, 4, Non-interruptible

Proposal 4181 - Observation 4 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Observation	Proposal 4181, Observation 4: Epoch 2 -- MIRI Spectroscopy Diagnostic Status: Warning Observing Template: MIRI Low Resolution Spectroscopy								Mon Jul 01 23:00:46 GMT 2024
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Generic Targets	#	Name	Criteria	Description					
	(1)	KILONOVA	LVK gravitational wave source believed to contain at least one neutron star (i.e., NS-NS or NS-BH) with a confirmed optical or infrared counterpart brighter than m~23 AB mag						
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	2 TACQ-STAR	F560W	FAST	10	1	1	27.75	146079.33
Template	Subarray			Obtain Verification Image?					
	FULL			false					
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps	
	1	ALONG SLIT NOD							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	100	20	40	1	2	11205.612	146079.36

Proposal 4181 - Observation 4 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Special Requirements	On Hold Awaiting GW trigger, to be scheduled with Epoch 2 NIRSPEC spectroscopy, both are ~5 days after the Epoch 1 observations Group Observations 3, 4, Non-interruptible
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Proposal 4181 - Observation 5 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Observation	Proposal 4181, Observation 5: Epoch 3 -- NIRSPEC low-res spectroscopy										Mon Jul 01 23:00:46 GMT 2024	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria		Description							
	(1)	KILONOVA	LVK gravitational wave source believed to contain at least one neutron star (i.e., NS-NS or NS-BH) with a confirmed optical or infrared counterpart brighter than m~23 AB mag									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	2 TACQ-STAR	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	146079.2	
Template	Slit				Subarray							
	S200A1				FULL							
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	100	1	1	NONE	4	4	5893.912	146079.37

Proposal 4181 - Observation 5 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Special Requirements	On Hold Awaiting GW trigger, to be scheduled ~10 days after the Epoch 2 observations 5 After 3 by 8 Days to 12 Days Group Observations 5, 6, Non-interruptible
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Proposal 4181 - Observation 6 - Infrared Spectroscopy of a Neutron Star Merger with JWST

Observation	Proposal 4181, Observation 6: Epoch 3 -- MIRI Imaging										Mon Jul 01 23:00:46 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria		Description						
	(1)	KILONOVA	LVK gravitational wave source believed to contain at least one neutron star (i.e., NS-NS or NS-BH) with a confirmed optical or infrared counterpart brighter than m~23 AB mag								
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	100	2	1	Dither 1	4	8	2231.132	146079.15
	2	F770W	FASTR1	150	4	1	Dither 1	4	16	6693.396	146079.22
Special Requirements	On Hold Awaiting GW trigger, to observed along wiht NIRSPEC low-res prism, roughly 10 days after the Epoch 2 observations										
	Group Observations 5, 6, Non-interruptible										